

Eric Wu

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Education

University of Waterloo

B.ASc in Mechatronics Engineering

Sep. 2025 – Apr. 2030 (Expected)

Waterloo, ON

Experience

Electromechanical Engineering Intern

May 2026 – August 2026

Spyder Controls

Lacombe, AB

- Designed and laser-cut **in-circuit test (ICT)** fixtures for PCBs in **CorelDRAW** using pogo pins and acetal, including hand-soldering and crimping wiring harnesses for **GD&T**-toleranced pin-to-pad contact.
- Root-caused PCBA fixture failures via **CAN bus** analysis and solder rework, restoring 5 fixtures to production.
- Engineered and deployed a self-aligning spring-loaded PCB nest at contract manufacturers, speeding verification to up to 400+ boards/day.
- Implemented **copper shielding** between neighbouring pogo-pin channels in high-density test fixtures, reducing false failures and cross-talk by 90%.
- Evaluated 6+ current-sensing components for accuracy, thermal behaviour, and charge/discharge performance using **weighted decision matrixes**.

Student Design Teams

Humanoid Robotics Designer

Apr. 2026 – Present

WATonomous

Waterloo, ON

- Designed and prototyped simplified CNC linkage mechanisms for a bipedal humanoid leg in **SolidWorks & Onshape**, reducing part count by 66%.
- Evaluated potential bearings and spacers for leg joint assemblies, assessing load capacity, RPM limits, and fit tolerances to meet the humanoid's dynamic walking requirements.
- Developed a **ROS 2** autonomous navigation system integrating **LiDAR**, obstacle inflation, and **A* path planning**.

Suspension Team Member

Sep. 2025 – Present

University of Waterloo Formula Electric

Waterloo, ON

- Machined 20+ suspension components to a ± 0.05 mm tolerance for the Formula SAE car.
- Modelled 3D-printed potentiometer mounts in **SolidWorks** for the front damper system, reducing operational vibration.
- Operated **manual mills and lathes** to machine aluminium and steel plates, L-brackets, spacers, rod plugs and top hats, with precision drilling and countersinking for flush-mounted hardware.

Projects

Omni-Assist | Autonomous Mobility Device

Apr. 2026

1st Place Hardware & Best Use of Viam, YHack @ Yale University

- Built a low-cost electromechanical drivetrain that turns chairs into self-driving, voice-controlled wheelchairs.
- Designed and 3D-printed ratcheting leg clamps, motor mounts, spacers, and wheel hubs in **SolidWorks** within 24 hours, mounting them to a frame built from aluminium extrusion and scrap metal.
- Wired the complete electrical system, including battery power, voltage regulation, and motor drivers, to drive four DC gear motors via an **Arduino**.
- Integrated **LiDAR** navigation and object detection for real-time obstacle avoidance.

Sensor-Based Arcade Machine | Automated 2-Player Ball-Toss Game

Nov. 2025

1st Place Game Track, BoxBots Hackathon

- Built a 4m-long, 2-player ball-toss arcade game in a 24-hour hackathon, integrating automated scoring and ball return into a fully functional electromechanical system.
- Engineered custom sensor and motor mounts in **SolidWorks**, integrating DC motors into an automated ball-return mechanism.
- Developed leader-follower **I²C** communication between 2 **Arduinos** for distributed sensing and real-time scoring, improving accuracy by 35% through calibration, debouncing, and timing-based filtering.

Skills

ME SolidWorks (CSWA Certified), Fusion 360, AutoCAD, Onshape, CorelDRAW, GD&T, Lathe, Mill, 3D Printing

EE Altium, ICT Testing, Prototyping & Debugging, Soldering & Harnessing, Sensors, Power Electronics, CANbus, Oscilloscopes

SWE/Robotics C++, Python, ROS 2, Linux, Git, STM32, Arduino, I²C, LiDAR